



SYDNEY GRAMMAR SCHOOL
MATHEMATICS DEPARTMENT
YEARLY EXAMINATIONS 2005

FORM V

MATHEMATICS

Examination date

Friday 14th October 2005

Time allowed

2 hours

Instructions

- All nine questions may be attempted.
- All nine questions are of equal value.
- All necessary working must be shown.
- Marks may not be awarded for careless or badly arranged work.
- Approved calculators and templates may be used.

Collection

- Write your name, class and master clearly on each booklet.
- Hand in the nine questions in a single well-ordered pile.
- Hand in a booklet for each question, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Keep the printed examination paper and bring it to your next Mathematics lesson.

5P: PKH 5Q: REN 5R: DS

Checklist

- SGS booklets: 9 per boy. A total of 500 booklets should be sufficient.
- Candidature: 37 boys.

Examiner

REN

QUESTION ONE (12 marks) Use a separate writing booklet.

Marks

(a) Simplify:

(i) $3\sqrt{3} - 7\sqrt{3}$

1

(ii) $4\sqrt{20}$

1

(iii) $\frac{\sqrt{28}}{\sqrt{7}}$

1

(b) (i) Solve the inequation $4x + 5 < 11$.

1

(ii) Graph the solution on the number line.

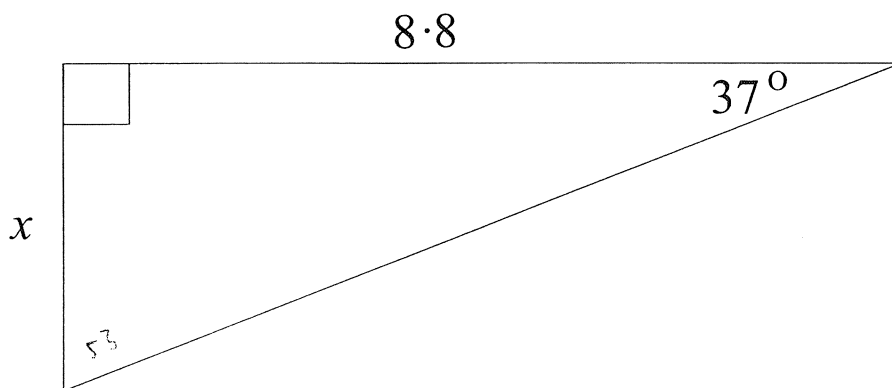
1

(c) Write down the exact value of $\cos 150^\circ$.

1

(d)

2



For the diagram above, find the value of x correct to 2 significant figures.

(e) Evaluate $16^{\frac{3}{4}}$.

1

(f) (i) If $2^x = 32$, find x .

1

(ii) Simplify $(a^3)^4 \div a^{12}$.

2

QUESTION TWO (12 marks) Use a separate writing booklet.

Marks

- (a) A series is given by $3 + 11 + 19 + 27 + \dots$
- (i) Show that the series is arithmetic. 1
 - (ii) Find the 200th term. 1
 - (iii) Find the sum of the first 200 terms. 1
 - (iv) Show that the n th term, T_n , is given by $T_n = 8n - 5$. 1
 - (v) Hence find the first term of the series that is greater than 1000. 2
- (b) A geometric series is given by $7 + 14 + 28 + 56 + \dots$
- (i) Find the 25th term of the series. 1
 - (ii) Find the sum of the first 20 terms of the series. 1
- (c) The 10th term of an arithmetic series can be written as $a + 9d$, where a is the first term and d is the common difference.
- (i) Write a similar expression for the 15th term of the series. 1
 - (ii) If the 10th term of the series is 16 and the 15th term is -14 , find a and d . 3

QUESTION THREE (12 marks) Use a separate writing booklet.

Marks

- (a) Differentiate the following:
- (i) $y = 5x^7$ 1
 - (ii) $y = 2x^{-6}$ 1
 - (iii) $y = 10x^{\frac{1}{2}}$ 1
 - (iv) $y = (4x - 7)^{10}$ 1
- (b) (i) Use the formula $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to differentiate $f(x) = 2x^2$ from first principles. 3
- (ii) Find the gradient of the curve $f(x) = 2x^2$ at the point where $x = -1$. 1
- (c) (i) State the product rule for differentiation. 1
- (ii) Use the rule to differentiate $y = 2x(x + 1)^9$. 1
(There is no need to fully factorise your answer.)
- (d) Differentiate $y = \frac{3x + 4}{x - 2}$. 2

QUESTION FOUR (12 marks) Use a separate writing booklet.

Marks

(a) Find the perpendicular distance from the point $(2, -1)$ to the line $3x + 4y - 6 = 0$.

2

(b) Solve the quadratic equation $x^2 - 5x - 7 = 0$.
(Write your solutions correct to 2 decimal places.)

2

(c) Solve the following for x :

(i) $\log_{27} x = \frac{1}{3}$

1

(ii) $\log_x 100\,000 = 5$

1

(d) Simplify the following:

(i) $\log_a 1$

1

(ii) $\log_a a^6$

1

(iii) $\log_a \sqrt{a}$

1

(e) Use the change of base formula to calculate $\log_7 100$ correct to 1 decimal place.

1

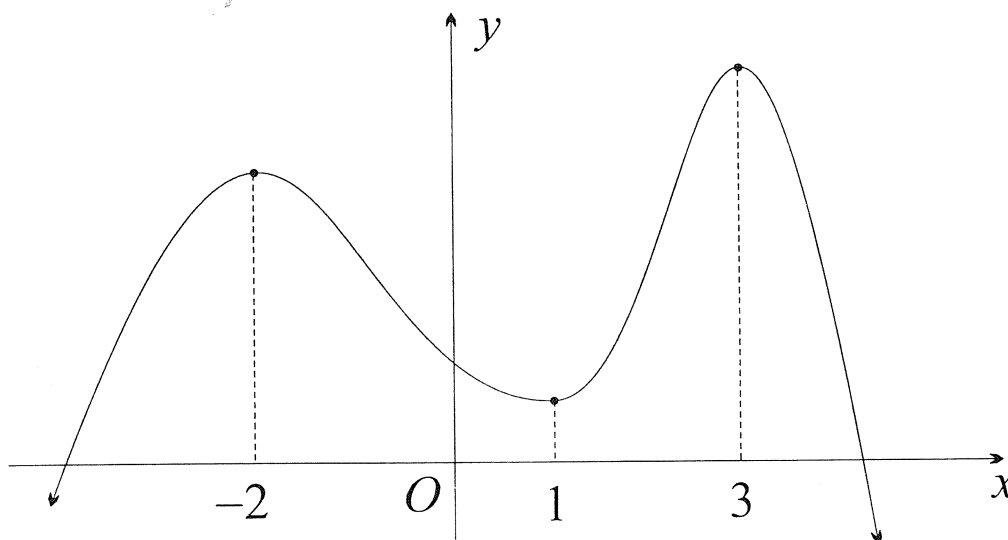
(f) Solve $3^x = 55$, giving your solution correct to 2 decimal places.

2

QUESTION FIVE (12 marks) Use a separate writing booklet.

Marks

(a)

The diagram above shows the graph of $y = f(x)$.(i) Write down the values of x for which $f(x)$ is stationary.

1

(ii) Write down the values of x for which $f(x)$ is(α) increasing

1

(β) decreasing

1

(b) The equation of a curve is given by $y = x^3 - 3x^2 + 5$.

(i) Show that $\frac{dy}{dx} = 3x(x - 2)$. 1

(ii) Find the equation of the tangent to the curve $y = x^3 - 3x^2 + 5$ at the point $(1, 3)$. 2

(iii) Find the value(s) of x for which the tangent is horizontal. 1

(c) Solve $x^4 - 7x^2 + 12 = 0$. 3

(d) (i) Find the discriminant of the quadratic equation $2x^2 - 7x + 2 = 0$. 1

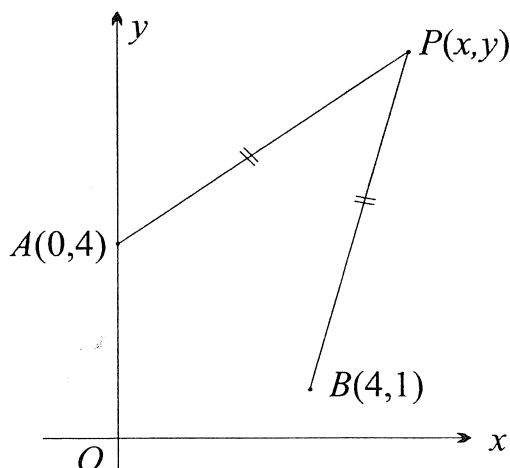
(ii) Explain why the quadratic equation has real roots. 1

QUESTION SIX (12 marks) Use a separate writing booklet.

Marks

(a) Find the centre and radius of the circle $x^2 + y^2 - 2x + 8y + 8 = 0$. 2

(b)



The diagram above shows the points $A(0, 4)$ and $B(4, 1)$ with $P(x, y)$ equidistant from A and B .

(i) Find the equation of the locus of the point P . 3

(ii) Describe the locus in words. 1

(c) The equation of a parabola is given by $x^2 = 16y$.

(i) Write down the co-ordinates of the focus. 1

(ii) Write down the equation of the directrix. 1

(d) A parabola has a focus at $(-1, -1)$ and directrix $y = 3$.

(i) On a number plane diagram sketch the graph of the parabola with the given focus and directrix. (Mark the focus, vertex and directrix on your diagram.) 2

(ii) Write down the co-ordinates of the vertex. 1

(iii) Write down the equation of the parabola. 1

QUESTION SEVEN (12 marks) Use a separate writing booklet.**Marks**

- (a) (i) Write down the formula for the limiting sum of a geometric series (assuming the limiting sum exists). **1**
- (ii) A geometric series is given by $27 + 18 + 12 + 8 + \dots$
- (α) Show that the series has a limiting sum. **1**
- (β) Find the limiting sum. **1**
- (b) By writing the recurring decimal $0.\dot{1}\dot{8}$ as an infinite series, express $0.\dot{1}\dot{8}$ as a fraction in simplest form. **2**
- (c) Let α and β be the roots of the quadratic equation $2x^2 - 4x + 1 = 0$. Without finding α and β , evaluate the following:
- (i) $\alpha + \beta$ **1**
- (ii) $\alpha\beta$ **1**
- (iii) $4\alpha^2\beta + 4\alpha\beta^2$ **1**
- (iv) $\alpha^2 + \beta^2$ **2**
- (d) Find the values of x for which the function $y = 3x^2 - 8x + 1$ is increasing. **2**

QUESTION EIGHT (12 marks) Use a separate writing booklet.**Marks**

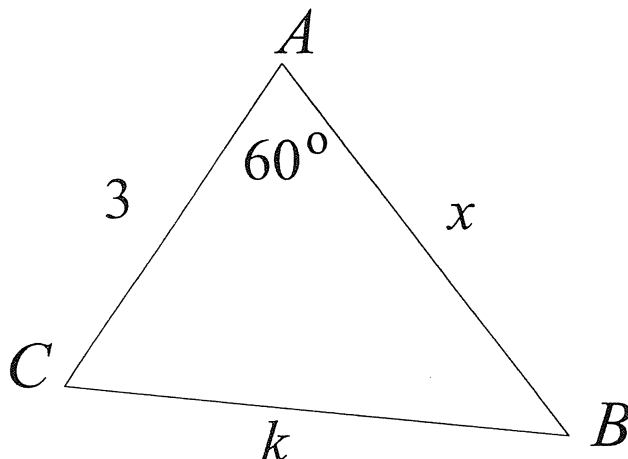
- (a) (i) Find the vertex and focus of the parabola $y^2 - 4y - 8x - 20 = 0$. **2**
- (ii) Sketch the parabola, showing the vertex and focus. **1**
- (b) Find the values of k for which the quadratic expression $x^2 - (k + 1)x + (k + 1)$ is positive definite. **3**
- (c) (i) If $f(x) = \sqrt{2x^2 + 1}$, show that $f'(x) = \frac{2x}{\sqrt{2x^2 + 1}}$. **1**
- (ii) Hence find the equation of the normal to the curve $y = \sqrt{2x^2 + 1}$ at the point $(2, 3)$. **2**
- (d) If the function $y = ax^2 + bx - 5$ has a stationary point at $(2, 1)$, find a and b . **3**

QUESTION NINE (12 marks) Use a separate writing booklet.

Marks

- (a) Find the point(s) on the curve $y = \frac{1}{x}$ where the gradient is equal to $-\frac{1}{4}$. 3

(b)



The diagram above shows $\triangle ABC$ with $AC = 3$, $AB = x$, $BC = k$ and $\angle BAC = 60^\circ$.

- (i) Use the cosine rule to show that $k^2 = x^2 - 3x + 9$. 2
- (ii) Explain why the length of side BC must be at least $\frac{3\sqrt{3}}{2}$. 2
- (c) An arithmetic series has first term a and common difference d .
- (i) Write down algebraic expressions for T_n and T_{n+1} . 1
- (ii) If the sum of the first n terms of the series is equal to twice the sum of the next n terms, show that $d = \frac{2a}{1 - 5n}$. 4

END OF EXAMINATION

FORM 5 MATHEMATICS SOLUTIONS

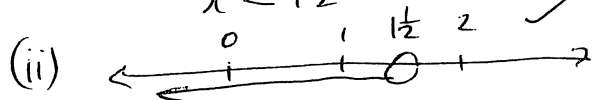
Q1

(a) (i) $3\sqrt{3} - 7\sqrt{3} = -4\sqrt{3}$ ✓

(ii) $4\sqrt{20} = 8\sqrt{5}$ ✓

(iii) $\frac{\sqrt{28}}{\sqrt{7}} = \sqrt{4} = 2$ ✓

(b) (i) $4x < 6$ ✓
 $x < 1\frac{1}{2}$ ✓



(c) $\cos 150^\circ = -\frac{\sqrt{3}}{2}$ ✓

(d) $\tan 37^\circ = \frac{x}{8.8}$ ✓

$x = 8.8 \times \tan 37^\circ$ ✓

$x \approx 6.6$ ✓

(e) $16^{\frac{3}{4}} = 8$ ✓

(f) (i) $2^x = 32$ ✓
 $x = 5$ ✓

(ii) $a^{12} \div a^{12} = 1$ ✓✓

(iv) $T_n = 3 + (n-1)8$
 $= 3 + 8n - 8$ ✓
 $= 8n - 5$ ✓

(v) $8n - 5 > 1000$ ✓
 $8n > 1005$
 $n > 125\frac{5}{8}$
 $n = 126$

$T_{126} = 8 \times 126 - 5$ ✓
 $= 1003$

(b) (i) $T_{25} = 7 \times 2^{24}$ ✓
 $= 117\,440\,512$

(ii) $S_{20} = \frac{7(2^{20}-1)}{2-1}$ ✓
 $= 7\,340\,025$

(c) (i) $a + 14d$ ✓

(ii) $a + 9d = 16$ ✓
 $a + 14d = -14$ ✓
 $5d = -30$ ✓
 $d = -6$ ✓

$a = 16 + 54 = 70$

$a = 70$ ✓

$d = -6$ ✓

Q2

(a) (i) $11 - 3 = 19 - 11 = 8$
 AP with $d = 8$ ✓

(ii) $T_{200} = 3 + 199 \times 8$ ✓
 $= 1595$

(iii) $S_{200} = \frac{200}{2} (3 + 1595)$ ✓
 $= 159\,800$ ✓

Q3

(a) (i) $y' = 35x^6$ ✓

(ii) $y' = -12x^{-7}$ ✓

(iii) $y' = 5x^{-\frac{1}{2}}$ ✓

(iv) $y' = 40(4x-7)^9$ ✓

(b) (i) $f'(x) = \lim_{h \rightarrow 0} \frac{2(x+h)^2 - 2x^2}{h}$ ✓

$= \lim_{h \rightarrow 0} \frac{2x^2 + 4xh + 2h^2 - 2x^2}{h}$

$= \lim_{h \rightarrow 0} \frac{h(4x + 2h)}{h}$

$= \lim_{h \rightarrow 0} 4x + 2h$ ✓

$= 4x$ ✓

(ii) $f'(-1) = 4x(-1) = -4$
gradient = -4 ✓

(c) (i) $y' = vu' + uv'$ ✓

(ii) $u = 2x$ $v = (x+1)^9$

$u' = 2$ $v' = 9(x+1)^8$

$y' = 2(x+1)^9 + 18x(x+1)^8$ ✓

(d) $u = 3x+4$ $v = x-2$
 $u' = 3$ $v' = 1$

$y' = \frac{vu' - uv'}{v^2}$

$= \frac{3(x-2) - 1 \times (3x+4)}{(x-2)^2}$ ✓

$= \frac{3x - 6 - 3x - 4}{(x-2)^2}$

$= -\frac{10}{(x-2)^2}$ ✓

Q4

(a) $d = \left| \frac{3 \times 2 + 4 \times (-1) - 6}{\sqrt{3^2 + 4^2}} \right|$
 $= \left| \frac{6 - 4 - 6}{5} \right|$ ✓
 $= \frac{4}{5}$ ✓

(b) $x = \frac{5 \pm \sqrt{(-5)^2 - 4 \times 1 \times (-7)}}{2}$ ✓
 $= \frac{5 \pm \sqrt{53}}{2}$
 $\approx 6.14 \text{ or } -1.14$ ✓

(c) (i) $x = 27^{\frac{1}{3}} = 3$ ✓

(ii) $x^5 = 100000$
 $x = 10$ ✓

(d) (i) $\log_a 1 = 0$ ✓

(ii) $\log_a a^6 = 6$ ✓

(iii) $\log_a a^{\frac{1}{2}} = \frac{1}{2}$ ✓

(e) $\log_7 100 = \frac{\log_{10} 100}{\log_{10} 7}$
 ≈ 2.4 ✓

(f) $x = \log_3 55$ ✓
 $= \frac{\log_{10} 55}{\log_{10} 3}$
 ≈ 3.65 ✓

Q5

- (a) (i) $x = -2, x = 1$ or $x = 3$ ✓
(ii) (a) $x < -2$ or $1 < x < 3$ ✓
(b) $-2 < x < 1$ or $x > 3$ ✓

(b) (i) $\frac{dy}{dx} = 3x^2 - 6x = 3x(x-2)$ ✓

(ii) At $(1, 3)$, $\frac{dy}{dx} = 3 \times 1 \times (1-2)$
 $= -3$ ✓

Eqn of tangent: $y - 3 = -3(x - 1)$
 $y - 3 = -3x + 3$
 $y = -3x + 6$ ✓

(iii) tangent horiz when $3x(x-2) = 0$
 $x = 0$ or $x = 2$ ✓

(c) let $a = x^2$ ✓
 $a^2 - 7a + 12 = 0$
 $(a-4)(a-3) = 0$ ✓
 $a = 4$ or 3
 $x = \pm 2$ or $\pm \sqrt{3}$ ✓

(d) (i) $\Delta = (-7)^2 - 4 \times 2 \times 2$
 $= 33$ ✓

(ii) Since $\Delta \geq 0$ equation has real roots. ✓

Q6

(a) $x^2 - 2x + 1 + y^2 + 8y + 16 = -8 + 17$
 $(x-1)^2 + (y+4)^2 = 9$ ✓

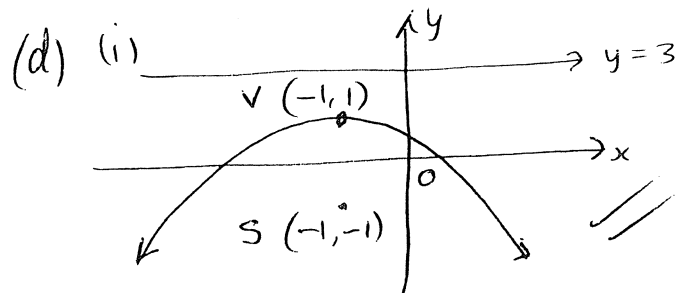
Centre $(1, -4)$ }
Radius = 3 } ✓

(b) (i) $\frac{AP}{AP^2} = \frac{BP}{BP^2}$ ✓

$(x-0)^2 + (y-4)^2 = (x-4)^2 + (y-1)^2$
 $x^2 + y^2 - 8y + 16 = x^2 - 8x + 16 + y^2 - 2y + 1$
 $8x - 6y - 1 = 0$ ✓

(ii) Perp. bisector of AB (accept "straight line") ✓

(c) (i) $(0, 4)$ ✓
(ii) $y = -4$ ✓



(ii) Vertex: $(-1, 1)$ ✓

(iii) Equation of parabola:
 $(x+1)^2 = -8(y-1)$ ✓

Q7

(a) (i) $S = \frac{a}{1-r}$ ✓

(ii) (a) $r = \frac{18}{27} = \frac{2}{3}$ ✓

Since $|\frac{2}{3}| < 1$ there is a limiting sum.

(b) $S = \frac{27}{1 - \frac{2}{3}}$ ✓
 $= 81$

(b) $0.\overline{18} = \frac{18}{100} + \frac{18}{10000} + \dots$

$r = \frac{1}{100}, a = \frac{18}{100}$

$S = \frac{\frac{18}{100}}{1 - \frac{1}{100}}$

$= \frac{18}{99}$

$= \frac{2}{11}$ ✓

(c) (i) $\alpha + \beta = 2$ ✓

(ii) $\alpha\beta = \frac{1}{2}$ ✓

(iii) $4\alpha\beta(\alpha + \beta) = 4 \times \frac{1}{2} \times 2$
 $= 4$ ✓

(iv) $(\alpha + \beta)^2 - 2\alpha\beta = 2^2 - 2 \times \frac{1}{2}$
 $= 3$ ✓

(d) $y' = 6x - 8$ ✓

Increasing when $6x - 8 > 0$
 $x > \frac{4}{3}$ ✓

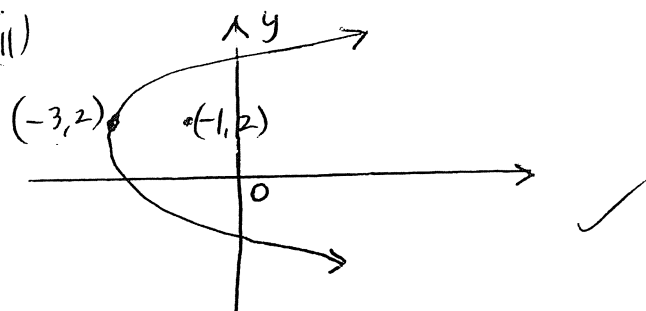
Q8

(a) (i) $y^2 - 4y + 4 = 8x + 24$
 $(y-2)^2 = 8(x+3)$ ✓

Vertex: $(-3, 2)$ }

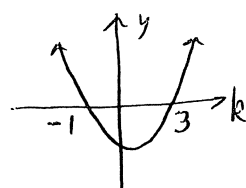
Focus: $(-1, 2)$ } ✓

(ii)



(b) $\Delta = [-(k+1)]^2 - 4 \times 1 \times (k+1)$
 $= k^2 + 2k + 1 - 4k - 4$
 $= k^2 - 2k - 3$ ✓

For pos. def. $k^2 - 2k - 3 < 0$ ✓
 $(k-3)(k+1) < 0$



$-1 < k < 3$ ✓

(c) (i) $f(x) = (2x^2 + 1)^{\frac{1}{2}}$
 $f'(x) = \frac{1}{2}(2x^2 + 1)^{-\frac{1}{2}} \times 4x$
 $= \frac{2x}{\sqrt{2x^2 + 1}}$ ✓

(ii) $f'(2) = \frac{2 \times 2}{\sqrt{2 \times 2^2 + 1}} = \frac{4}{3}$
 grad of normal $= -\frac{3}{4}$ ✓

Eqn of normal: $y - 3 = -\frac{3}{4}(x - 2)$
 $4y - 12 = -3x + 6$
 $3x + 4y - 18 = 0$ ✓

(d) $y' = 2ax + b$

At $(2, 1)$, $y' = 0$

$4a + b = 0 \dots (1)$ ✓

Also, $1 = 4a + 2b - 5$ (since $(2, 1)$ on curve)

$4a + 2b = 6$

$2a + b = 3 \dots (2)$ ✓

(1) - (2): $2a = -3$

$a = -\frac{3}{2}$

$b = 6$ ✓

Q9

$$(a) \quad y' = -\frac{1}{x^2} \quad \checkmark$$
$$-\frac{1}{x^2} = -\frac{1}{4}$$
$$x^2 = 4$$
$$x = 2 \text{ or } -2 \quad \checkmark$$

when $x = 2$, $y = \frac{1}{2}$

when $x = -2$, $y = -\frac{1}{2}$

points : $(2, \frac{1}{2})$ $\checkmark$
 $(-2, -\frac{1}{2})$

$$(b) \quad (i) \quad k^2 = 3^2 + x^2 - 2 \times 3 \times x \times \cos 60^\circ \quad \checkmark$$
$$k^2 = 9 + x^2 - 2 \times 3 \times x \times \frac{1}{2} \quad \checkmark$$
$$k^2 = x^2 + 9 - 3x = x^2 - 3x + 9$$

$$(ii) \quad x^2 - 3x + 9 - k^2 = 0$$
$$\Delta = (-3)^2 - 4 \times 1 \times (9 - k^2)$$
$$= 9 - 36 + 4k^2 \quad \checkmark$$
$$= 4k^2 - 27$$

For real roots, $4k^2 - 27 \geq 0$

$$k^2 \geq \frac{27}{4}$$

$$k \geq \frac{\sqrt{27}}{2} \quad (k > 0)$$

$$k \geq \frac{3\sqrt{3}}{2} \quad \checkmark$$

$$(c) \quad (i) \quad T_n = a + (n-1)d \quad \checkmark$$

$$T_{n+1} = a + nd$$

$$(ii) \quad S_n = \frac{n}{2} [2a + (n-1)d] \quad (1) \quad \checkmark$$

(Sum of first n terms)

For sum of next n terms :

first term = $a + nd$

$$\text{Sum} = \frac{n}{2} [2(a + nd) + (n-1)d] \quad \checkmark$$

$$= \frac{n}{2} [2a + 2nd + nd - d] \quad \checkmark$$

$$= \frac{n}{2} (2a + 3nd - d) \quad (2)$$

Now, $S_n = 2 \times \text{Sum} \quad \checkmark$

$$\frac{n}{2} (2a + nd - d) = n(2a + 3nd - d) \quad \checkmark$$

$$2a + nd - d = 2(2a + 3nd - d)$$

$$2a + nd - d = 4a + 6nd - 2d$$

$$d - 5nd = 2a$$

$$d(1 - 5n) = 2a \quad \checkmark$$

$$d = \frac{2a}{1 - 5n}$$